Mr. Partington's iGCSE Computer Science Revision Notes

Paper 1

Exam Technique

Know what the examiners can ask you questions about

Have a look at the IGCSE Computer Science syllabus that you have been studying and are going to be taking the exam for. The list of topics will make a great checklist for your revision. If you find something that you don't understand or haven't made any notes about, then find out about it. *Anything* that is mentioned in the syllabus could be used in an examination question.

Read and understand examination questions

What are you being asked to do?

1. Read the question
2. Understand the type of instruction you are being given: Complete, Describe, Draw, Explain, Give and State all require different actions.
3. If the question makes use of a specific scenario or context then make sure that all of your answers are relevant to that context. For example if the question is about security measures for an offline device, then using a internet based firewall would not be appropriate!
4. Decide on the information required but remember that you are sitting an iGCSE examination and most answers will require more than just a single word. If you have finished well before the time allotted, you may well have fallen into this trap.
5. Always use correct technical terms and avoid the use of trade names. For example, talk about the use of an operating system rather than the use of 'Windows 10'.
6. Decide how much information is required to fulfill the number of marks available and if in doubt, add more!

Help the examiner help you!

- Make sure your answers are easy to read (if in doubt, write it again, clearer!).
- Read through the entire question before you start to answer it, give yourself thinking time and decide how you will format your answer before writing.
- Make it easy for the examiner to see where he/she should give you the marks. This also helps you make sure that you will can gain every mark available.
- Answer every question! There is no point leaving blank spaces, you will not lose marks for incorrect answers, so you may as well have a guess.

1.1 - Data Representation

1.1.1 Binary systems

a) Recognise the use of binary numbers in computer systems

Binary is the **base 2** number system. It is used in computer systems because computers store data with the use of of switches that are in two states: on or off, 1 or 0.

b) Convert positive denary integers into binary and positive binary integers into denary (a maximum of 16 bits will be used)

To convert binary (base 2) to denary (base 10) we use the following table. Adding up all of the numbers where a '1' bit occurs.

8-bit example:
Original number: 01101010

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
64	32	16	8	4	3	2	1
0	1	1	0	1	0	1	0

= 32 + 16 + 4 + 2 = 54

To convert denary (base 10) to binary (base 2) we use the same table, placing a 1 in the correct positions until it adds up to the value we need.

c) show understanding of the concept of a byte and how the byte is used to measure memory size

Measurement of memory size

Name of memory size	Number of bits	Equivalent value
1 bit	1	1 bit
1 nibble	4	1 nibble
1 byte	8	1 byte
1 kilobyte (1 KB)	2^{10}	1 024 bytes
1 megabyte (1 MB)	2^{20}	1 048 576 bytes
1 gigabyte (1 GB)	2^{30}	1 073 741 824 bytes
1 terabyte (1 TB)	2^{40}	1 099 511 627 776 bytes
1 petabyte (1 PB)	2^{50}	1 125 899 906 842 624 bytes

d) Use binary in computer registers for a given application

A register is a place to store a group of bits.
8 bit registers are often used to control electronic devices. Each bit is used to control a specific value. For example:

1	0	1	0	0	1	1	0
Motor A on	Motor A off	Motor B on	Motor B off	A direction forward	A direction backward	B direction forward	B direction backward

This means that motor A and B are both on. Motor B is rotating forward and motor A is rotating backwards.

1.1.2 Hexadecimal

a) represent positive numbers in hexadecimal notation

Hexadecimal is a base 16 number system that uses the digits 0 to 9 and the letters A to F to represent each hexadecimal digit.
Denary, Binary, Hexadecimal

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**
0000	**0001**	**0010**	**0011**	**0100**	**0101**	**0110**	**0111**
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**

8	**9**	**10**	**11**	**12**	**13**	**14**	**15**
1000	**1001**	**1010**	**1011**	**1100**	**1101**	**1110**	**1111**
8	**9**	**A**	**B**	**C**	**D**	**E**	**F**

b) show understanding of the reasons for choosing hexadecimal notation to represent numbers

Hexadecimal:

- Is easier/quicker to read than binary
- Is easier/quicker to write/type than binary
- Takes up less room on the screen than binary
- Is easier to debug than binary
- Is less prone to human error than binary

c) convert positive hexadecimal integers to and from denary (a maximum of four hexadecimal digits will be required)

To convert hexadecimal integers to and from denary it is often easier to go through binary. To start, write out each hexadecimal digit as 4 binary bits. For example: 4AE would become 0100 1010 1110. Then convert the number to denary as described in 1.1.1 (b).
0100 1010 1110 = 1024 + 128 + 32 + 8 + 4 + 2 = 1198.

d) represent numbers stored in registers and main memory as hexadecimal

A register is a place to store a group of bits. We can use our knowledge of how to convert between base 2, base 10 and base 16 to represent numbers stored in registers and main memory as hexadecimal.

e) identify current uses of hexadecimal numbers in computing, such as defining colours in Hypertext Markup Language (HTML), Media Access Control (MAC) addresses, assembly languages and machine code, debugging

The hexadecimal system is used for:

1. **Memory Dumps** - useful when developing new software, this allows the contents of the memory to be seen by the writer, thus enabling errors to be detected; also used in diagnostics when a computer malfunctions, hex is used since it is easier to use than a long string of binary values.
2. **HTML Colours** - hypertext markup language uses 6 digit hexadecimal codes to represent colours. The first two digits represent the amount of red, the next two represent green and the last two represent the amount of blue.
3. **MAC Address** - See 1.2.2 e)
4. **Web addresses** - ASCII code can be used to replace the URL or parts of the URL.
5. **Assembly code/machine code** - using hex can make it easier and faster to write and less error-prone than writing code in binary.

1.1.3 Data Storage

a) show understanding that sound (music), pictures, video, text and numbers are stored in different formats

Examples of various file formats are shown below:

Sound	.aif, .iff, .m4a, .mid, .mp3, .wav, .wma
Pictures	.bmp, .gif, .jpg, .png, .psd, .tiff
Video	.3gp, .avi, .flv, .m4v, .mov, .mp4, .mpg, .vob, .wmv
Text	.doc, .docx, .odt, .pages, .rtf, .txt, .wpd
Numbers	.xls, .xls, .db, .sheets

b) **identify and describe methods of error detection and correction, such as parity checks, check digits, checksums and Automatic Repeat reQuests (ARQ)**

Parity checks
Parity checks can be **even** (check for an even number of 1-bits) or **odd** (check for an odd number of 1-bits). The bit that is added to make the even or odd total is known as the **parity bit.** It is possible that **two** errors in the data could result in an error not being detected.

Automatic Repeat Request (ARQ)
ARQ uses an acknowledgement to indicate data has been received correctly. A timeout is used which is the time allowed to elapse before an acknowledgement is received. If an acknowledgement is not received before the timeout, then the data is sent again.

Checksum
Data is sent in blocks. An additional value, called the checksum, is sent at the end of the block of data. The checksum is calculated based on the number of bytes in the block of data. If the checksum calculated at the receiver's end doesn't match the checksum that is sent at the end of the data block, then an error has occurred.

Echo Check
Data is transmitted. The data is then returned to the sender. The sender compares the data sent with the data received back. If they are different then an error has occurred and the data needs to be sent again.

c) **show understanding of the concept of Musical Instrument Digital Interface (MIDI) files, JPEG files, MP3 and MP4 files**

MIDI (Musical Instrument Digital Interface)
This system is associated with the storage of music files. No sounds are stored (as in WAV, MP3 or MP4) but instead, digital signals are sent using a protocol that allows musical instruments to interact. Examples of signals might include: NOTE 64 On, NOTE 64 OFF, NOTE 70 On Velocity 50, PITCH BEND -50%.
Since MIDI files don't contain actual audio tracks, their size is relatively small. This makes them good for devices with small storage capacity, for example: storing ringtones on a mobile phone.

MP3
MP3 uses lossy audio compression to store music in an MP3 file format. These can often be up to 90% smaller than comparable CD music files. The music quality is retained by removing sounds that the human ear often can't hear.
MP4 files are slightly different as they can often store video or photos and not just audio.

Text and number file formats
Text is stored in ASCII format and text files are usually stored in a lossless format.
Numbers can be stored as real, integer, currency, and so on. Lossless format is used since accuracy of data is very important.

Joint Photographic Experts Group (JPEG) files
JPEG files use a lossy format file compression method. JPEG is used to store photographs as a reduced file size. They rely on certain properties of the human eye (e.g. its inability to detect small brightness differences or colour hues).

d) **show understanding of the principles of data compression (lossless and lossy) applied to music/ video, photos and text files**

Lossless and lossy file formats
With lossless file compression, all the data from the original files is reconstructed when the file is uncompressed. There is no loss in quality or loss in data when using lossless compression.
With lossy file compression, unnecessary data is eliminated forming a file and it can't be reconstructed to get back to the original file. For example: sending a large image file might be unnecessary for the purpose you need it for, so the image is compressed first to make the file take less time to upload and send.

1.2 - Communication and Internet Technologies

1.2.1 Data Transmission

a) **show understanding of what is meant by transmission of data**

Data transmission is where digital data is transferred from one device or component to another. Data transmission can be wireless (wifi, bluetooth, NFC, 3g/4G/5G etc.) , or send along a physical medium, such as a copper network cable, fibre optic cable etc.

b) **distinguish between serial and parallel data transmission**

Serial data transmission is where data is sent **one bit at a time** over a **single wire** or channel.
Parallel data transmission is where data is sent **several bits at a time** over **multiple wires** or channels.

c) **distinguish between simplex, duplex and half-duplex data transmission**

Simplex data transmission is where data is sent in **one direction only**.
Half-duplex data transmission is where data is sent in **two directions, but not at the same time**.
Duplex data transmission is where data is sent in **two directions at the same time**.

d) **show understanding of the reasons for choosing serial or parallel data transmission**

Serial data transmission is slower than parallel, but is good for long distances as the data cannot become skewed.
Parallel data transmission is faster than serial, but data can become skewed over long distances (so is more suitable for short distances)

e) **show understanding of the need to check for errors**

Data transmission is susceptible to interference. This can be electrical, environmental, physical, magnetic etc. Interference will cause data to become skewed and not arrive in the correct order, or for some data to not arrive at all. This is why we must check that a data transmission was successful using an error checking method.

f) **explain how parity bits are used for error detection**

A parity bit is added to data to give it even or odd parity. It is usually added at the beginning or the end of the data, or each byte of data. See 1.1.3 b).

g) **show understanding of the use of serial and parallel data transmission, in Universal Serial Bus (USB) and Integrated Circuit (IC)**

As its name suggests, USB uses serial data transmission, an IC normally uses parallel data transmission.

1.2.2 Security Aspects

a) **show understanding of the security aspects of using the Internet and understand what methods are available to help minimise the risks**

Firewalls
Firewalls can help stop unauthorised access from the internet by examining the traffic between a user's computer or local network and a public network (e.g. the internet).
A firewall:
- Monitors incoming and outgoing traffic on a network
- Checks whether incoming and outgoing traffic meet certain criteria, if data fails criteria, the firewall blocks the traffic and warns the user
- can log all incoming and outgoing traffic
- Criteria can be set to prevent access to certain websites, this can be done by the firewall keeping a list of all undesirable IP addresses (a blacklist)
- Warns the user if software tries to access an external data source

b) **show understanding of the Internet risks associated with malware, including viruses, spyware and hacking**

Hacking

Hacking is a method of gaining unauthorised (and sometimes illegal) access to a computer system. This can lead to identity theft and loss or corruption of data. The risk of hacking can be minimised by using strong passwords and firewalls.

Cracking

Cracking is the editing of a program source code so that it can be exploited or changed for a specific purpose (usually without the owner's consent, and therefore illegal). It is often done for malicious purposes, e.g. removing password protection or removing the need for a licence to run the software. It is often difficult to guard against; software engineers need to make it difficult to identify ‘back doors’ (ways of breaking into the software).

Viruses

A virus is program code that can replicate or copy itself with the intention of deleting or corrupting data or files, or causing the computer to malfunction in another way. They can cause the computer to run slow (due to the hard disk filling up with data, or the processor being directed to other tasks) or crash (due to missing critical files). The risk of viruses can be minimised by running anti-virus software or not opening emails or software from unknown sources.

Phishing

Legitimate-looking emails are sent to users; on opening the email, the user could be asked to supply personal or financial details or they may be asked to click on a link which sends them to a fake website where the user could be asked to supply personal data. Many email providers try to filter out phishing emails and are largely successful, but some still slip through the filter. There are a number of signs to look out for in phishing emails:

- Poor spelling and grammar
- A unexpected message asks for personal information
- You did not initiate the action or communication
- A message that makes unrealistic threats e.g. sending something to everyone in your contact list.
- A message that makes unrealistic promises e.g. cash prizes, lottery winning etc.

Pharming

Malicious code that is installed on a users computer or web server. The code redirects the use to a fake website without their knowledge (it may look exactly like the genuine website and even appear to have the correct URL). Once the user is sent to the fake website they may be asked to give out personal or financial data.
Some anti-spyware or antivirus software can identify and remove pharming code on a computer. The user could also look for clues such as lack of encryption, strange looking URL, computer running slow etc.

Wardriving

This is the act of locating and using wireless internet connections illegally. It can lead to stealing of internet time and bandwidth, also other user's passwords and other data. It can be prevented with the use of strong wireless network security (e.g WEP).

Spyware/key logging software

This gathers data by monitoring key presses on user's keyboards and sending the data back to the person who sent the spyware. Sends important data, such as passwords back to the originator of the spyware. Prevented by use of anti-spyware or antivirus software or the use of a mouse or drop down box to enter passwords rather than a keyboard.

c) **explain how anti-virus and other protection software helps to protect the user from security risks**

Anti-virus software can scan for and detect viruses, quarantine viruses and delete viruses from a computer system. It can be set to automatically scan new files, and periodically scan an entire system. It will warn the user using an interrupt if it finds a virus. A **false positive** is when anti-virus software flags something as a virus when it is not a virus.

Anti-spyware software can scan for and detect spyware. It will warn the user using an interrupt if it detects spyware. It can delete spyware from a computer system.

Auto-Update software will download and install updates for your software as and when they are available. It is good to keep programs updated as updates are often more secure (vulnerabilities have been identified and fixed), more features have been added, or compatibility may have been improved.

1.2.3 Internet Principles of Operation

a) **show understanding of the role of the browser**

An **internet browser** is software which allows the user to display a web page on the computer screen. The software interprets/translates the HTML from the website and shows the result.

b) **show understanding of the role of an Internet Service Provider (ISP)**

An **ISP** is company that provides the user with access to the internet; they usually charge a monthly fee. The ISP gives the user an account and sometimes provides the user with an email address.

c) **show understanding of what is meant by hypertext transfer protocol (http and https) and HTML**

Hypertext Transfer Protocol **(http)** is a set of rules that must be obeyed when transferring files across the internet, this may include web pages or individual files. Http**s** indicates that the protocol is using SSL encryption to send files or web pages securely.

d) **distinguish between HTML structure and presentation**

HTML is used when writing and developing web pages. HTML uses <tags> to bracket pieces of codes. Hexadecimal values are used to represent different colours in HTML. In HTML some tags are used to define **structure** (these do not make much difference to the front-end of the website, for example <head> and <body> tags). Other tags are used to define **presentation** (these are used to define how items are displayed on the page, for example <b> would make text appear **bold.**

e) **show understanding of the concepts of MAC address, Internet Protocol (IP) address, Uniform Resource Locator (URL) and cookies**

MAC address
a media access control address is used to uniquely identify a device on a network. The first 6 hex digits represent the manufacturer and the latter 6 are unique to that specific device. MAC addresses can be UAA (a universally administered address) or LAA (a locally administered address).

IP (Internet Protocol) address
Each device is given an IP address as soon as it connects to the internet. The IP address will be different each time the device connects.

URL (Uniform Resource Locator)
A URL is the address that you type in to access a website. It is made up of different parts, for example:

https://www.google.co.uk/index.html

Protocol   Hostname or Domain Name   File name

Cookies
Pieces of data which allow detection of web pages viewed by a user and to store their preferences. Cookies can be deleted from the user's desktop although this can remove some of the features of certain websites.

1.3 Hardware and Software

1.3.1 Logic gates

a) **use logic gates to create electronic circuits**

A **logic gate** takes in binary inputs and produces a binary output.
Logic circuits are made up of several logic gates and are designed to carry out a specific function.

b) **understand and define the functions of NOT, AND, OR, NAND, NOR and XOR (EOR) gates, including the binary output produced from all the possible binary inputs (all gates, except the NOT gate, will have 2 inputs only)**

See below (d)

c) draw truth tables and recognise a logic gate from its truth table

See below (d)

d) recognise and use the standard symbols used to represent logic gates.

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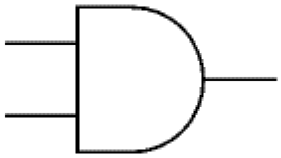
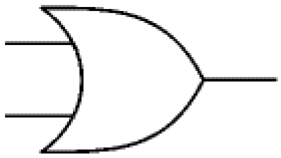
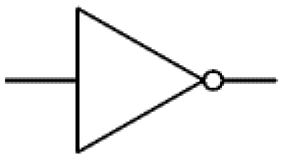
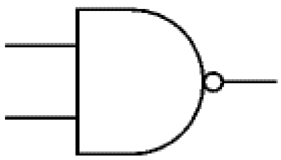
<tr><th>Type of GATE</th><th>Diagram</th><th>Truth Table</th></tr>
<tr><td>AND</td><td></td><td><table><tr><th>A</th><th>B</th><th>A ∧ B</th></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table></td></tr>
<tr><td>OR</td><td></td><td><table><tr><th>A</th><th>B</th><th>A V B</th></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table></td></tr>
<tr><td>NOT</td><td></td><td><table><tr><th>A</th><th>¬A</th></tr><tr><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td></tr></table></td></tr>
<tr><td>NAND</td><td></td><td><table><tr><th>A</th><th>B</th><th>¬(A ∧ B)</th></tr><tr><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td></tr></table></td></tr>
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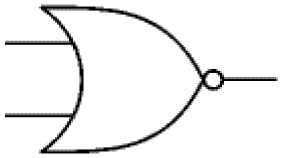
<td></td>
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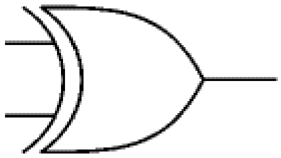
<td></td>
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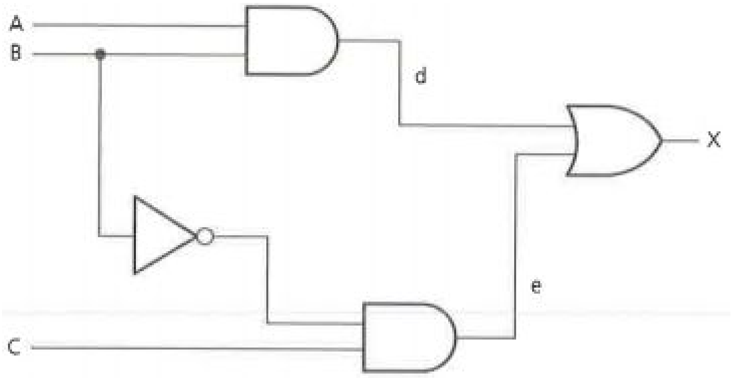
e) Produce truth tables for given logic circuits.

If you are asked to do this you will be given a space to show your working. It is a good idea to annotate the circuit adding in some locations where you will work out the intermediate answers before you evaluate your overall answer.

An example might be: The student has added in locations d and e to help them work out the overall answer.



With the truth table:

Inputs			Working		Output
A	B	C	D	E	X
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	0	1

f) Produce a logic circuit to solve a given problem or to implement a given written logic statement.

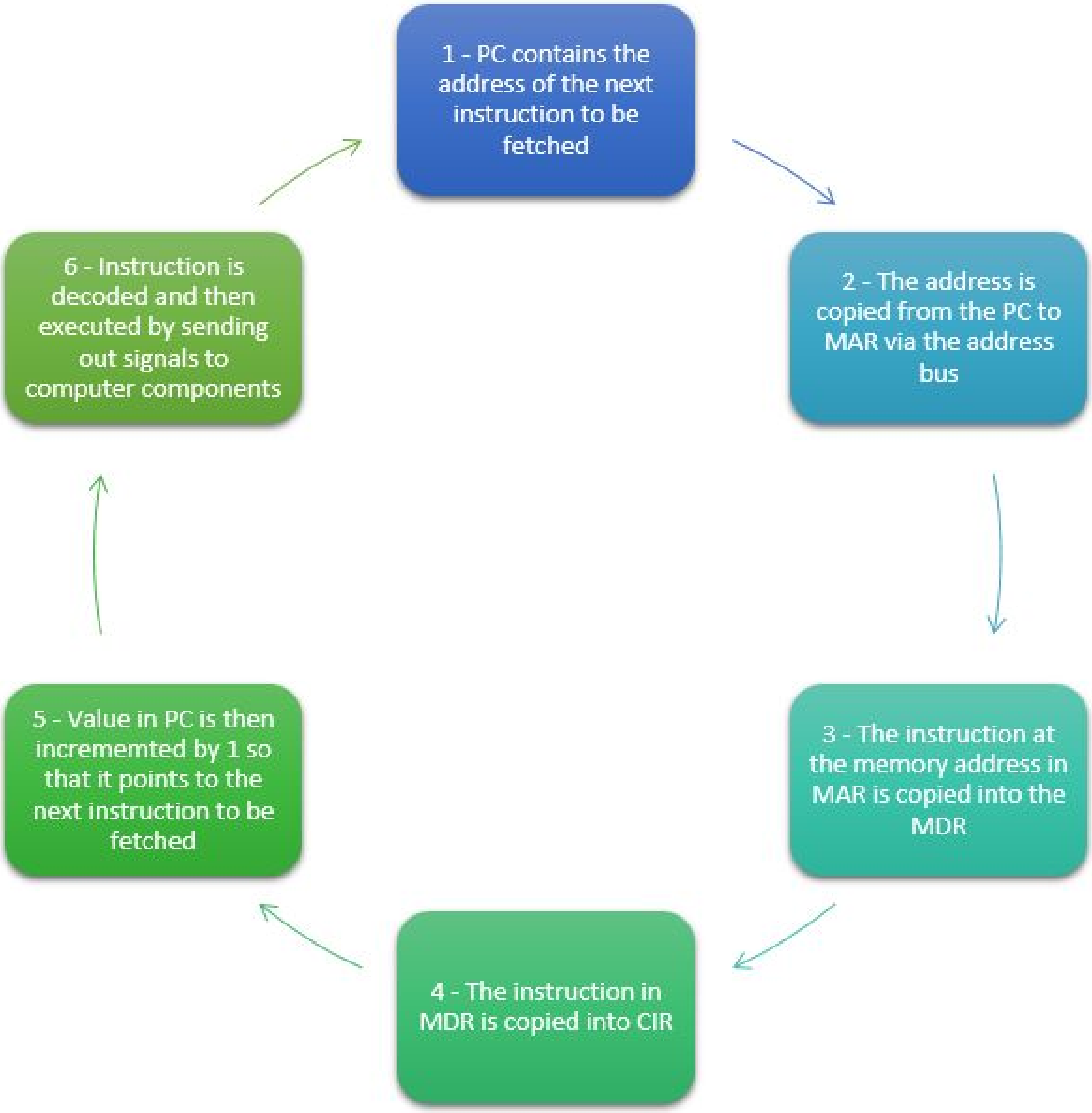
This is perhaps one of the hardest things you will be asked to do. It is useful to annotate the question (particularly if it is quite wordy) highlighting the key words such as AND, OR, NOT. It is also worth annotating what value is meant by the wording in the question (for example it might say “if the temperature is over 25 degrees” in logic terms, this means something like T=1).

1.3.2 Computer architecture and the fetch-execute cycle

a) **show understanding of the basic Von Neumann model for a computer system and the stored program concept (program instructions and data are stored in main memory and instructions are fetched and executed one after another)**

The Von Neumann model holds both programs and data in the memory. Data moves between the memory unit and the processor.
A register is a high-speed storage area within the processor. All data must be in a register before it can be processed. There are 5 registered used in the Von Neumann model: Memory Address Register (**MAR**), the Memory Data Register (**MDR**), the Accumulator (**ACC**), the Program Counter (**PC**) and the Current Instruction Register (**CIR**).
A bus is the connection used to move data around the processor and also send out control signals to synchronise the internal operations. There are three different buses used: **Address bus** (used to carry signals relating to addresses from the processor to the memory, it is unidirectional), **Data bus** (this sends data between the processor, memory uni and the input and output devices, it is bi-directional) and the **Control bus** (this carried signals relating to control and coordination of all activities within the computer; it can be unidirectional or bidirectional depending on what it is connecting).

b) **describe the stages of the fetch-execute cycle, including the use of registers and buses**



1.3.3 Input devices

a) **describe the principles of operation (how each device works) of these input devices: 2D and 3D scanners, barcode readers, Quick Response (QR) code readers, digital cameras, keyboards, mice, touch screens, interactive whiteboards, microphones**

2D Scanners

These convert hard-copy documents into an digital format which can be stored in a computer memory. Scanners operate by using a scan head which moves across the document shining a bright light which reflects and produces an image which is sent to a lens via a series of mirrors. Applications of 2D scanner include: scanning passports at airports, scanning newspapers for archives, scanning textbooks for digital reading, scanning worksheets to complete online.

3D Scanners

3D scanners scan solid objects and produce an electronic 3D image. They use light or lasers and measure their reflections to calculate the dimensions of an object. 3D scanners produce a 3D model of a solid object.

Barcode Readers
Barcodes consist of a series of dark and light lines of varying thickness. Each digit or character is represented by a number of lines. A barcode reader uses light or a laser to detect the series of lines and input this as a series of characters into the computer.
Advantages of using barcode readers include:

- Allows for automatic stock control
- Allows for faster checkouts in shops
- Can check customer's buying habits and customise offers based on trends
- Less chance of errors at tills

QR (quick response) codes
QR codes are a type of barcode. However, they can hold considerably more data. QR codes are usually read by built-in cameras in smartphones or tablets using an app, but can be read by conventional barcode scanners.

Digital Cameras
Digital cameras contain a microprocessor which automatically:

- Adjust shutter speed
- Focus the lense(s)
- Operates the flash
- Adjusts the aperture
- Removes 'red eye'
- Reduce hand shake

Etc.
Images are captured when light passes through the lens onto a light-sensitive cell, which is made up of thousands of tiny elements to capture each pixel.

Keyboards/keypads
Keyboards are the most common input device. Keys are pressed by the operator to enter data directly into the computer. When a key is pressed, it completes a circuit and a signal is sent to the microprocessor which interprets which key has been pressed.

Pointing Devices
The most common pointing devices are the mouse and the trackpad. They are used to control a cursor on a screen or to select options from menus. Other pointing devices include: interactive whiteboards and graphics tablets.

Microphone
Microphones are used to input sound into a computer. When the microphone picks up sound, a diaphragm vibrates producing an electric signal. A analogue to digital converter then converts the signal into digital values which can be processed by the computer. Voice and speech recognition systems both use microphones.

Touchscreens

Touchscreens allow selection to be made by simply touching an icon or menu option. They are also use in many devices to allow an input via a 'virtual keyboard'.
Mobile phones and tablets are some of the biggest users of touchscreen technology. The most common systems are capacitive, infra-red or resistive.

	Capacitive	**Infrared**	**Resistive**
How it works	- Uses layers of glass that act as a capacitor - When the top layer is touched the current changes - The microprocessor works out the coordinates of where the screen was touched.	- Uses glass and can either detect heat or uses infrared sensors to detect touch. - microprocessor works out where the screen was touched based on sensor/heat data	- uses upper layer of polyester and bottom layer of glass - when top layer touched, it completes a circuit - microprocessor works out coordinates of where the screen was touched
Benefits	- this is a medium-cost technology - good screen visibility in strong sunlight - allows multi-touch capability - very durable	- allows multi-touch capability - can use bare fingers, gloved hand or stylus - good screen durability	- relatively inexpensive technology - can use bare fingers, gloved hand or stylus
Drawbacks	- can only use bare fingers or a conductive stylus	- relatively expressive technology - heat-sensitive systems only allow bare fingers to be used	- poor visibility in strong sunlight - doesn't allow multi-touch capability - screen is vulnerable to scratches

b) describe how these principles are applied to real-life scenarios, for example: scanning of passports at airports, barcode readers at supermarket checkouts, and touch screens on mobile devices

Please see 1.3.3 a)

c) **describe how a range of sensors can be used to input data into a computer system, including light, temperature, magnetic field, gas, pressure, moisture, humidity, pH and motion**

Sensors
Sensors send data from the real world to a computer. They often require an analogue to digital converter (ADC) to change the data into a format that the computer (or microprocessor) can understand. Sensors form part of many monitoring or control systems.

d) **describe how these sensors are used in real-life scenarios, for example: street lights, security devices, pollution control, games, and household and industrial applications**

- Sensors continuously send a signal to the system
- The signal is sent through a analogue to digital converter
- The value is compared to a predefined value.
- If the value is outside the acceptable range then a signal is sent to an actuator or alarm
- If the value is within the acceptable range then the system continuously checks the incoming values

1.3.4 Output devices

a) **describe the principles of operation of the following output devices: inkjet, laser and 3D printers; 2D and 3D cutters; speakers and headphones; actuators; flat-panel display screens, such as Liquid Crystal Display (LCD) and Light-Emitting Diodes (LED) display; LCD projectors and Digital Light Projectors (DLP)**

Printers
The three most common types of printer are: laser, inkjet and dot matrix.

Type of Printer	How it works	Advantages	Disadvantages
Laser Printer	- Uses positive and negative charges on the print drum and paper. - Toner sticks to the paper where it is charged - Toner permanently fixed using a fuser	- high quality of printing - relatively inexpensive to buy printer - large toner cartridges and large paper trays - very fast printing of documents	- can be expensive to maintain - produce health hazards such as ozone or toner particles in the air
Inkjet	- Use liquid ink system which sprays ink onto the paper line by line as the paper moves through the printer. - The ink system uses either thermal bubble or piezoelectric	- very high quality of printing - suitable for printing photos - inexpensive to buy printer	- ink is expensive to buy and quickly runs out on a large print run - usually have small paper trays - can be noisy in operation compared

	technologies.		to laser printer
Dot Matrix Printer	- Use a matrix of pins which strike an inked ribbon to produce characters in a matrix on the paper.	- not adversely affected by damp or dusty atmospheres - allows use of multipart stationery (i.e. carbon copies) - allows use of continuous/fan-folde d stationery	- relatively expensive to buy printer - poor print quality - very noisy and very slow at printing

3D Printer
3D printers produce an actual solid objects. They are built up in the printer in a number of think layers. Can use a number of different materials: powdered resin, powdered metal, ceramic power, plastic reel, or even paper. 3D printer have been used to make prosthetic limbs, aerospace parts, fashion and art item and even to make parts that are no longer in production by conventional manufacturing.

2D/3D Cutters
These are used to cut into materials to form 3D objects and are controlled by computers and software (CAD/CAM). Common materials include glass, crystal, metal, plastics and wood.

Headphones/Loudspeakers
Headphones and loudspeakers convert analogue voltages into sound. If the output is form a computer, the digital signals are first converted into analogue voltages using a digital to analogue converter.

Actuators
An actuator is a component of a machine that is responsible for moving and controlling a mechanism or system, for example by opening a valve.

LCD and LED monitors
Modern flat screen monitors are usually either LCD or LED (sometimes OLED). LCD monitors use liquid crystal to generate the image and require a backlight as LCD does not produce its own light. LED and OLED screens use lots of light emitting diodes to generate coloured light that makes up a display.

Light Projectors
DLP (digital light projectors) use millions of micro mirrors which can switch on or off several thousand times a second creating various shades of light. Colour filters allow the shades of light to be converted into colours which are projected onto a large screen.
LCD projectors use chromatic-coated mirrors which reflect light at different wavelengths. The light components pass through three LCD screens which are then recombined using a prism to produce the colour image which is projected onto a large screen.

b) describe how these principles are applied to real-life scenarios, for example: printing single items on demand or in large volumes; use of small screens on mobile devices

Please see 1.3.4 a)

1.3.5 Memory, storage devices and media

a) show understanding of the difference between: primary, secondary and off-line storage and provide examples of each, such as: primary: Read Only Memory (ROM), and Random Access Memory (RAM) secondary: hard disk drive (HDD) and Solid State Drive (SSD); off-line: Digital Versatile Disc (DVD), Compact Disc (CD), Blu-ray disc, USB flash memory and removable HDD

Primary memory is memory that is directly accessed by the CPU, e.g. RAM, ROM.
Secondary memory is memory that is not directly accessed by the CPU, e.g. HDD, SSD and offline storage (including CDs, DVDs and blu-rays). **Off-line** storage is memory that can be safely removed from the computer.

RAM - Random Access Memory
This memory is volatile (data is lost when the power is off). It is used to store programs and files that are currently in use. For any program or file to be used it must be copied into the RAM first.

ROM - Read Only Memory
This memory is non-volatile. It is used to store the start-up procedures or BIOS (basic input output system). This memory can only be read and not written to.

b) describe the principles of operation of a range of types of storage device and media including magnetic, optical and solid state

Magnetic storage devices use a magnetised medium to store data. Whether a bit is magnetised or not indicated is that bit is a 1 or a 0. Examples include magnetic tape, standard HDDs (hard disk drives). They are generally cheap, but not that reliable as the system includes moving parts.

Optical storage devices use a laser to read/write data stored on a disk. Red lasers are used to read DVDs and CDs and a blue laser is used to read/write to a blu-ray disk. Bumps and pits are recorded onto a track, which represent 1s and 0s.Most disks have a single track that starts in the centre of the disk and spirals outwards, however, a DVD-RAM disk has tracks in concentric circles. CDs typically store 700MB, DVDs 4.7GB and Blu-Ray discs 25GB. Dual layer DVDs store 8.5GB and dual layer Blu-Ray disks store 50GB. Optical memory is susceptible to scratches on the surface on the disk which can skew data, as the disk has be spun round at high speeds it is also susceptible to mechanical failure due to moving parts.

Flash storage uses semiconductors to store data. Millions of transistors are either on or off which represent the 1s and 0s. Flash memory is solid. There are no moving parts, unlike magnetic and optical, this makes it more reliable. It is more expensive to manufacture than magnetic or optical storage. SSD (solid state drives) use flash memory, as do USB flash drives and SD cards.

c) **describe how these principles are applied to currently available storage solutions, such as SSDs, HDDs, USB flash memory, DVDs, CDs and Blu-ray discs**

See 1.3.5 b)

d) **calculate the storage requirement of a file**

Files are measured using the measurements in 1.1.1 c). If you are asked to calculate the storage requirements of a file you will be give all of the numbers you need in the question. They may ask you to give your answer in a different form to that in the question (for example they may give you a number of bits and ask you to work out the number of bytes). Image size is a popular choice for this sort of question, it is important that you include the colour depth when calculating this size ((height in pixels x width in pixels) x colour bit depth = total size). Be careful with units, and simplify calculations where possible.

1.3.6 Operating systems

a) **describe the purpose of an operating system (Candidates will be required to understand the purpose and function of an operating system and why it is needed. They will not be required to understand how operating systems work.)**

An **operating system** is the software running in the background of a computer. It manages many of the basic functions of the computer, including:

- Human-computer interface (HCI)
- Multi-tasking
- Multiprogramming
- Batch Processing
- Error handling/reporting
- Load/run applications
- Management of user accounts
- File utilities (such as copy and delete)
- Processor management
- Memory management
- Real-time processing
- Interrupt handling
- Security (log on, username/password etc.)
- Input and output control

Not every computer requires an operating system, for example the microprocessors controlling ovens or washing machines will not require one as they carry out simple, unchanging tasks.

b) **show understanding of the need for interrupts**

An **interrupt** is a signal sent from a device or software to the processor requesting its attention. The interrupt will cause the processor to temporarily stop what it is doing to service the interrupt. Examples include: pressing ALT+CTRL+BREAK, a paper jam in a printer, or software trying to divide by zero.

1.3.7 High- and low-level languages and their translators

a) show understanding of the need for both high-level and low-level languages

A computer program is a list of instructions that enable a computer to perform a specific task. Computer programs can be written in high-level or low-level language depending on the task to be performed. Low level languages access the processor and memory directly however high level languages require translating into a low level language before they can run. Most programs are written in high-level languages.
High-level languages enable a programmer to focus on the problem to be solved and require no knowledge of the hardware and instruction set of the processor that will be used. High-level languages can be used on different types of computer.
Low-level languages relate to the specific processor of a particular computer. Low-level languages can refer to machine code or assembly language (which needs to be translates into machine code).

b) show understanding of the need for compilers when translating programs written in a high-level language

A compiler is a computer program that translates a program written in a high-level language into machine code so that it can be directly used by a computer to perform a task. A compiler will attempt to translate the entire code at once, and report all of the errors at the end, at once.

c) show understanding of the use of interpreters with high-level language programs

An interpreter is a computer program that reads a statement from a program written in a high-level language, performs the action specified and the does the same with the next statement and so on. This means that any errors are reported as they occur, line by line. If an Interpreter encounters an error it will not move onto the next line.

d) show understanding of the need for assemblers when translating programs written in assembly language

An assembler is a computer program that translates a program written in an assembly language into machine code so that it can be directly used by a computer to perform a required task.

1.4 Security

1.4.1

a) show understanding of the need to keep data safe from accidental damage, including corruption and human errors

Data can be accidentally damaged by a number of external factors e.g. fire, accidental deletion, accidental overwriting, hardware failure, software faults etc.

b) **show understanding of the need to keep data safe from malicious actions, including unauthorised viewing, deleting, copying and corruption**

Malicious acts can also prevent data from being safe. Most accidental methods of data damage can be done intentionally.

1.4.2

a) **show understanding of how data are kept safe when stored and transmitted, including:**

i) **use of passwords, both entered at a keyboard and biometric**
Strong passwords are important. Weak passwords are easy to guess and use common, easy to remember words or letters.

ii) **use of firewalls, both software and hardware, including proxy servers**
Firewalls are described in 1.2.2

iii) **use of security protocols such as Secure Socket Layer (SSL) and Transport Layer Security (TLS)**
SSL is a protocol (set of rules) used by computers to communicate with each other across a network. It allows data to be send a received securely across a network, including the internet. HTTP**S** indicated that SSL is being used. When a user wants to access a website, the web browser asks the web server to identify itself, the web server sends a copy of the SSL certificate which the web browser authenticates. If it is OK then SSL data transfer beings between the user's computer and the web server.
TLS is similar to SSL but is more recent and more effective. Only recent web browsers support TLS. It uses a record protocol and a handshake protocol.

iv) **use of symmetric encryption (plain text, cypher text and use of a key) showing understanding that increasing the length of a key increases the strength of the encryption**
Encryption can be symmetric:
- Uses a secret key; when the key is applied, the plain text goes through an encryption algorithm to produce ciphertext. The recipient needs a key to then decrypt the message back into plain text.
- The main risk is that the sender and recipient need the same key, which is susceptible to being hacked or intercepted

Encryption can also be asymmetric:
- Asymmetric encryption uses public and private keys.
- The public key is available to everybody and the private key is only known to the user.
- Both keys are needed to encrypt and decrypt messages

1.4.3

a) **show understanding of the need to keep online systems safe from attacks including denial of service attacks, phishing, pharming**

A denial of service attack is an attempt at preventing users from accessing part of a network, usually internet servers. They can prevent users from accessing certain websites or accessing online services. This is achieved by the attacker flooding the network with useless traffic. For example, sending thousands of requests to a website or sending out thousands of spam emails to users 'clogging up a system'.

Other definitions can be found in 1.2.2

1.4.4

a) **describe how the knowledge from 1.4.1, 1.4.2 and 1.4.3 can be applied to real-life scenarios including, for example, online banking, shopping**

Make sure your answers to this sort of question relate directly to the context of the question. For example, don't mention internet security if the system in question is offline.

1.5 Ethics

a) **Show understanding of computer ethics, including copyright issues and plagiarism.**

Computer ethics is a set of principles set out to regulate the use of computers to stop unethical things from happening. Copyright is a system to prevent somebody from copying something without permission. Plagiarism is the act of copying somebody's work and claiming it is your own.

b) **distinguish between free software, freeware and shareware**

Free software - Users have the freedom to run, copy, change or adapt the software.
Freeware - Users can download free of charge, but it is subject to copyright laws so the user cannot copy, change or adapt the software.
Shareware - Users are allowed to try out shareware free for a trial period. At the end of this period, the user will be requested to pay a few. Sometimes the trial version does not have all of the features of the full version.

c) **show understanding of the ethical issues raised by the spread of electronic communication and computer systems, including hacking, cracking and production of malware**

Ethical issues include:

- privacy - should authorities be able to see our personal data?
- Digital ownership - copyright and open source
- Data gathering - CCTV and other monitoring methods
- Access costs - should the internet be free?
- Ethical hacking - is hacking ever good?